

genaut — frontier report

source CSV : genaut/logs/2statelap0acc/default.csv
corpus rows: 30

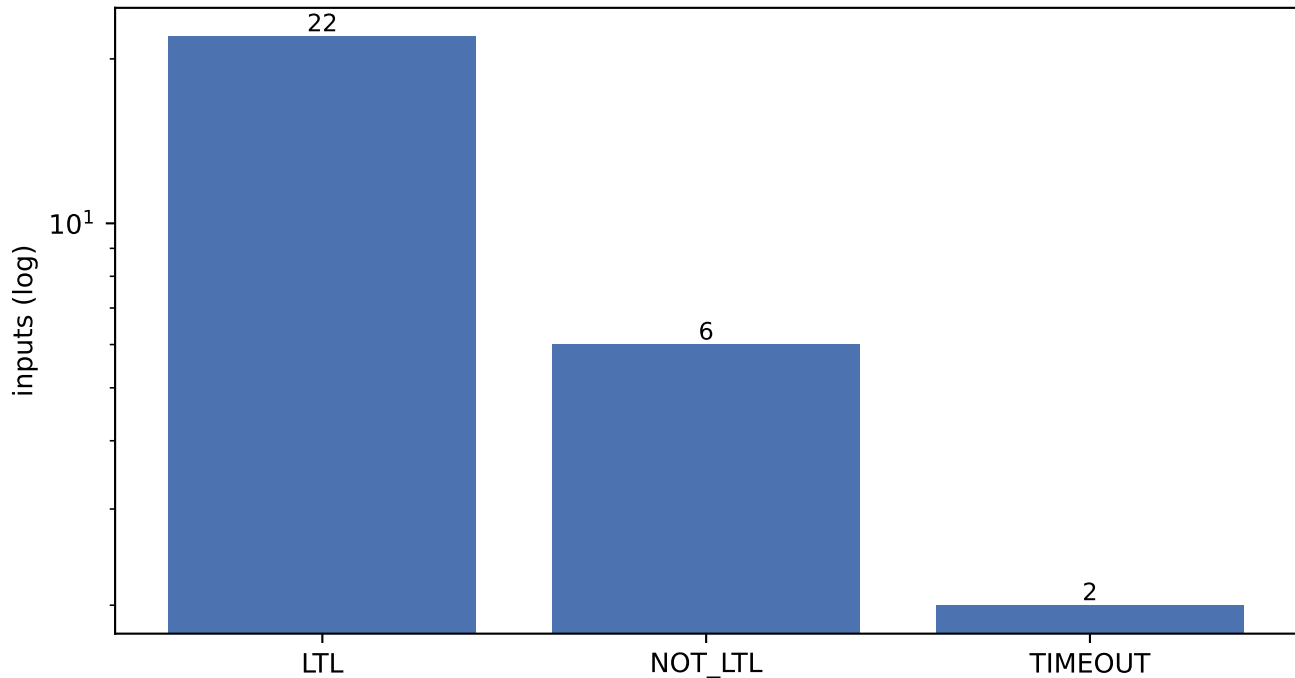
sections:

- Ventilation – LTL / not-LTL / ambiguous
- Formula collapse – structural (md5) concentration
- Size frontier – flattened formula size
- Routes to `true` – which technique built the universal 1
- Build time & timeouts
- Verification – spot-oracle validation

Ventilation — LTL / not-LTL / ambiguous

```
corpus inputs: 30
  LTL-definable (formula built) :    22  ( 73.3%)
    not-LTL (decided)           :     6  ( 20.0%)
    ambiguous (timeout, unknown) :     2  (  6.7%)
decided (LTL or not-LTL): 28  ( 93.3%)
```

Answer frontier (by result)



Formula collapse — structural (md5) concentration

LTL answers: 22 (0 with a gated/unprintable string)

distinct structures (md5): 17

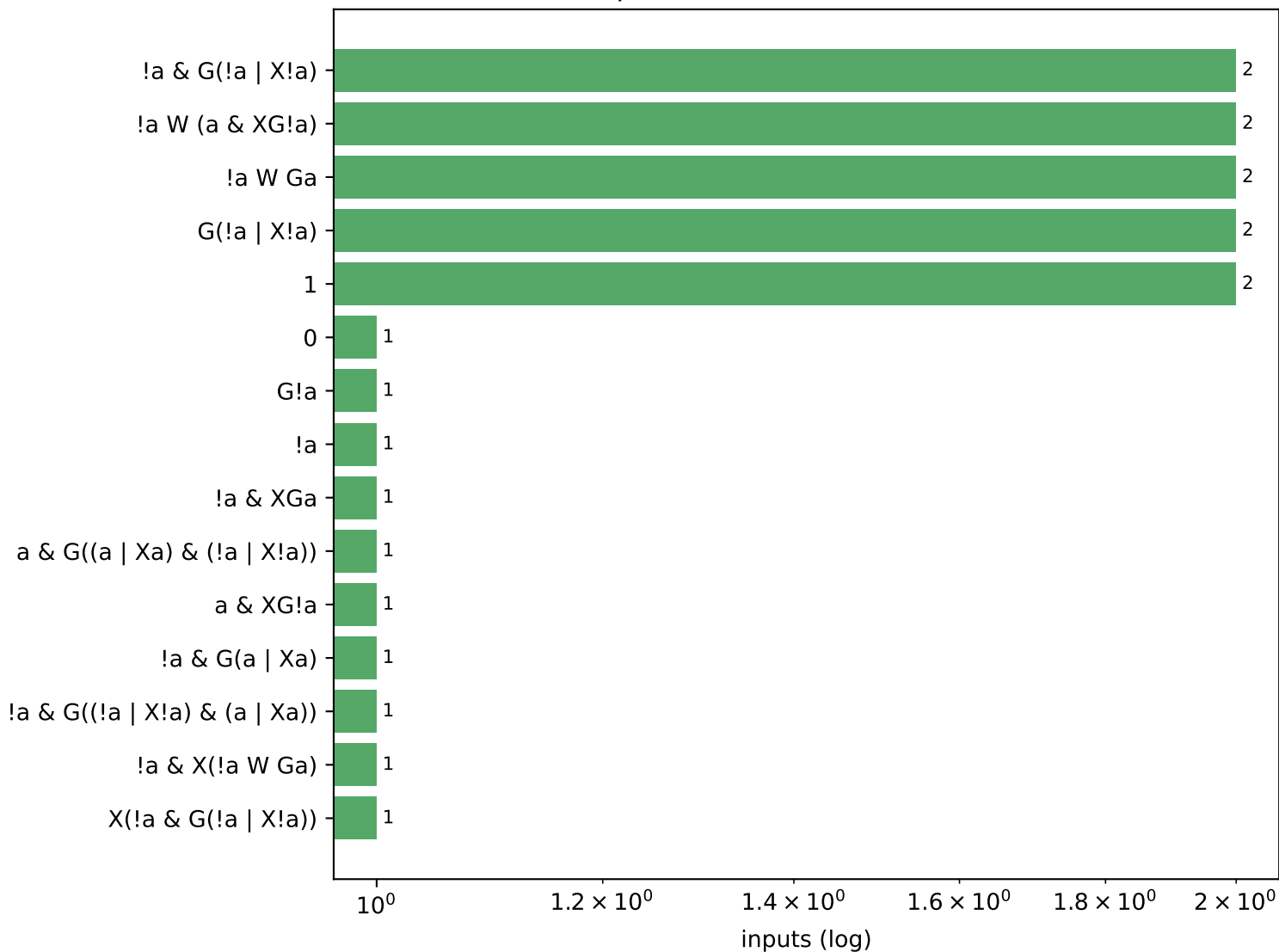
80% of answers covered by the top 13 structure(s)

90% of answers covered by the top 15 structure(s)

top 15 structures:

2	(	9.1%	!	a	&	G	(	!	a		X	!	a	)								
2	(	9.1%	!	a	W	(	a	&	XG	!	a	)										
2	(	9.1%	!	a	W	G	a															
2	(	9.1%	G	(	!	a		X	!	a	)											
2	(	9.1%	1																			
1	(	4.5%	0																			
1	(	4.5%	G	!	a																	
1	(	4.5%	!	a																		
1	(	4.5%	!	a	&	XG	a															
1	(	4.5%	a	&	G	(	(	a		Xa	)	&	(	!	a		X	!	a	)	)	
1	(	4.5%	a	&	XG	!	a															
1	(	4.5%	!	a	&	G	(	a		Xa	)											
1	(	4.5%	!	a	&	G	(	(	!	a		X	!	a	)	&	(	a		Xa	)	)
1	(	4.5%	!	a	&	X	(	!	a	W	G	a	)									
1	(	4.5%	X	(	!	a	&	G	(	!	a		X	!	a	)						

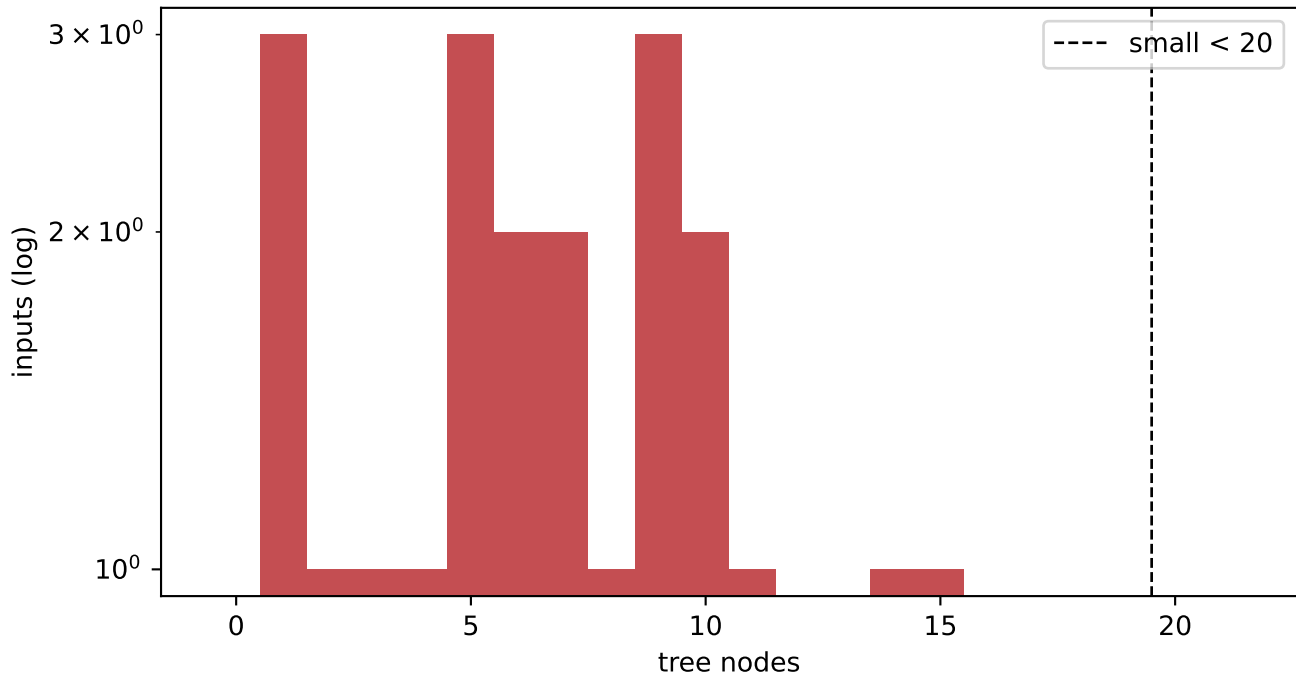
Top 15 reconstructed structures



Size frontier — flattened formula size

```
LTL answers with a size: 22
trivial (tree<=1, i.e. true/false): 3 ( 13.6%)
small (tree<20): 22 (100.0%)
larger (tree>=20): 0 ( 0.0%)
tree nodes: median 6 mean 6.7 max 15
```

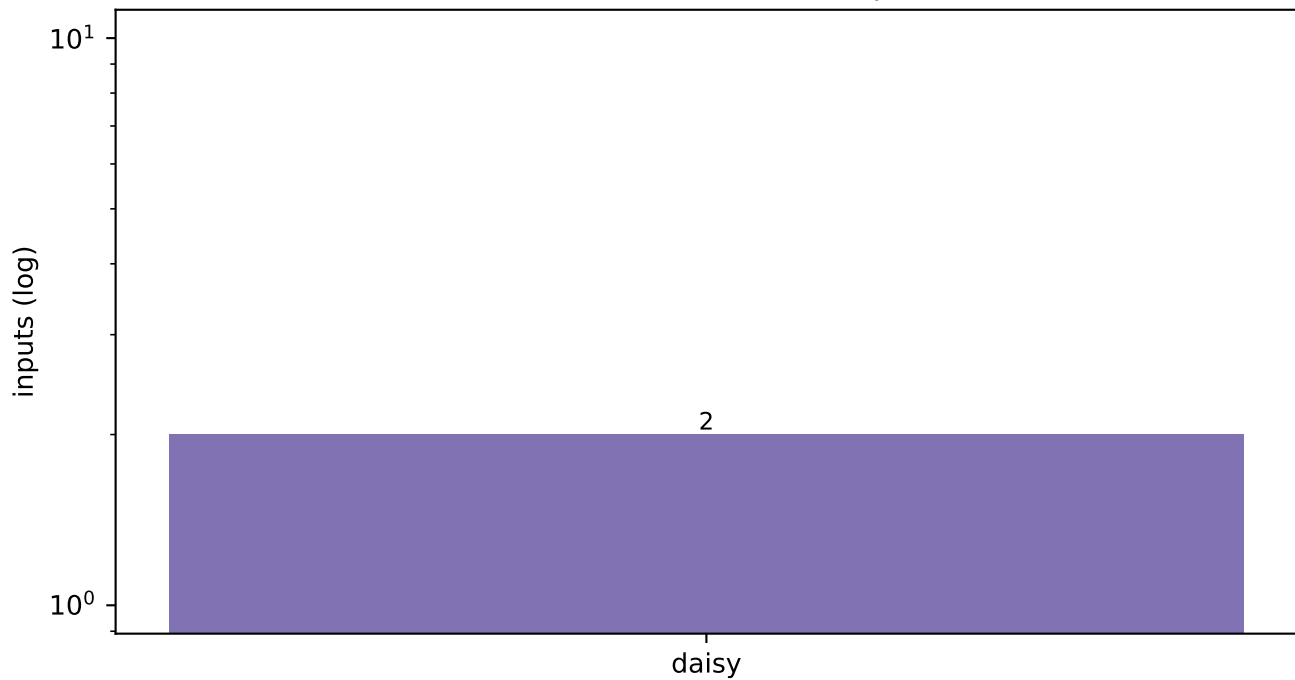
Flattened size (tree nodes); clipped at 21 (0 larger pooled at right)



Routes to `true` — which technique built the universal 1

```
universal answers (formula == '1'): 2  
  2 (100.0%) daisy  
build_s over true rows: total 0.1s  mean 0.070s  max 0.070s
```

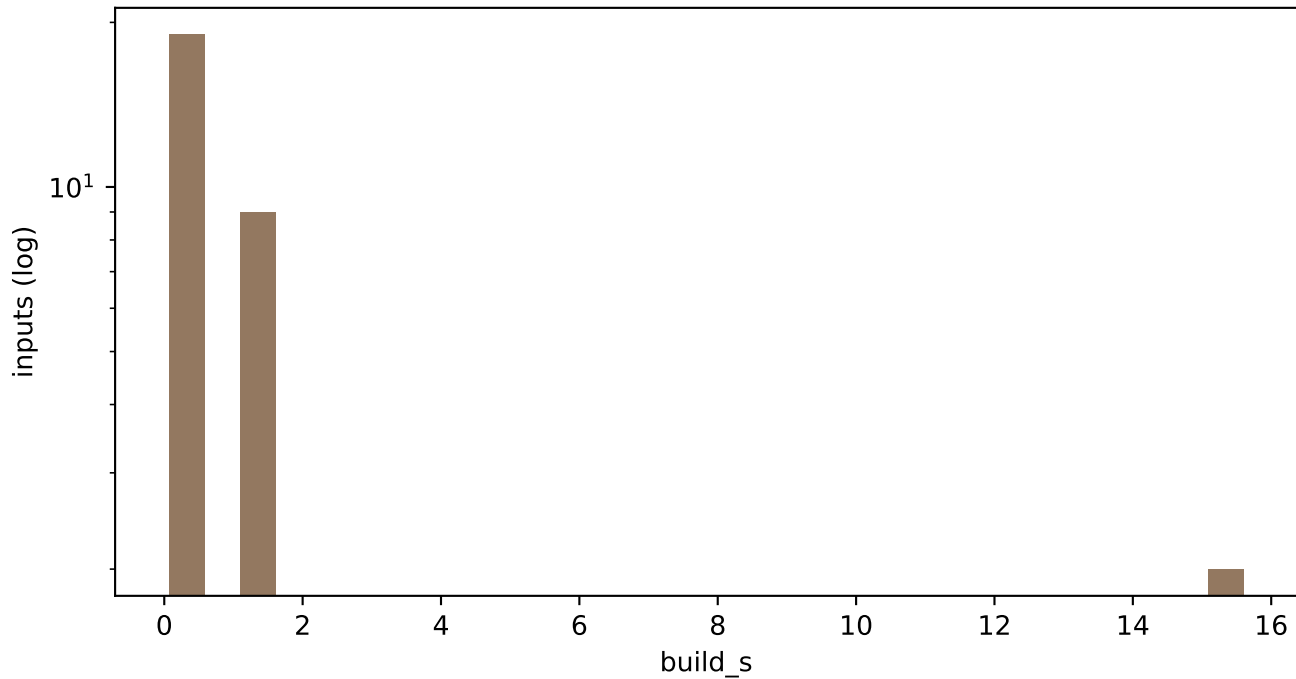

Routes to `true` (technique)



Build time & timeouts

```
total build: 42.5s   mean 1.417s   max 15.599s
  LTL           n= 22   mean 0.221s   max 1.233s
  NOT_LTL       n=  6   mean 1.119s   max 1.155s
timeouts: 2
  2statelap0acc_086.hoa
  2statelap0acc_106.hoa
```

Build-time distribution



Verification — spot-oracle validation

validation over 22 LTL answers:
True 22 (100.0%)
NOT-equivalent (FALSE): 0 CLEAN